CCNA DevNet with Python for Network Engineers(Upskill & Reskill)

Objective:

The program is designed to provide individuals with the knowledge, skills, and hands-on networking and coding experience necessary to excel in the field of Network Automation.

CCNA DevNet training module is helping developers and IT professionals learn, code, and innovate using Networking technologies and platforms. It focuses on providing resources, tools, and support to develop applications and solutions for networking and infrastructure products.

Here are some key features and objectives of the AimNxt CCNA DevNet with Python:

Comprehensive Networking and Development Training: The program offers a comprehensive curriculum that covers various Networking Hardware concepts, networking concepts, technologies, and Scripting, Programming skills .

Hands-on Lab Experience: AimNxt CCNA DevNet emphasizes practical learning through hands-on lab exercises. Participants have access to real-world applications and software tools to gain practical experience in advanced Network programming and automation.

Certification Preparation: The program prepares participants for industry-recognized certifications- CCNA DevNet.

Industry-Relevant Skills: The program focuses on developing skills that are in high demand in the industry.

Industry Experts and Mentors: The program offers access to industry experts and mentors who provide guidance and support to participants. These professionals share their knowledge and insights, offer career advice, and help participants navigate the IT industry effectively.

Career Development and Job Placement Assistance: The AimNxt CCNA DevNet assists participants in developing their careers in advanced Network programming and Network automation. It provides resources for resume building, interview preparation.

Overview

AimNxt CCNA DevNet is the perfect start if you know you want to build, manage and support of enterprise IT infrastructure.

CCNA DevNet can helps you prepare for a wide variety of IT jobs, including:

- Software Engineer
- Network Automation Engineer
- Network programmer
- Network Developer
- Software Developer

Duration:

Total- 45 Days (Including lab Hours).

Printed Material/LMS and Coding assistance.

Modules:

Chapter 1: Introduction to Cisco DevNet Associate Certification.

Chapter 2: Software Development and Design.

1. Software Development Lifecycle.
2. Common Design Patterns.
3. Linux BASH.
4. Software Version Control.
5. Git.

Chapter 3: Introduction to Python – Includes Lab.

1. Understanding Python Syntax.
2. Data Types and Variables.
3. Input and Output.
4. Flow Control with Conditionals and Loops.

Lab - Practice the effects of fundamental Python commands.
Lab - Program Python data structure.
Lab - Write scripts to demonstrate Python loops and conditional code.

Chapter 4: Python Functions, Classes, and Modules – Includes Lab.

1. Python Functions.
2. Using Arguments and Parameters.
3. Object-Oriented Programming and Python.
4. Python Classes.
5. Working with Python Modules.

Lab - Write Python Functions.
Lab - Write Python code of Modules and Packages.
Lab - Implement Python code in IDEs.
Lab - Debug and test Python code.

Chapter 5 Working with Data in Python – Includes Lab.

1. File Input and Output.
2. Parsing Data.
3. Comma-Separated Values (CSV).
4. JavaScript Object Notation (JSON).
5. Extensible Markup Language (XML).
6. YAML Ain't Markup Language (YAML).
7. Error Handling in Python.
8. Test-Driven Development.
9. Unit Testing.

Lab - Parse API Data Formats with Python.
Lab - Create Python scripts to interact with files.

Chapter 6 IOS CLI Automation using Python Built-in & 3rd Party Libraries – Includes Lab.

1. Understanding CLI Automation Basics.
2. Telnetlib Built-in Python Library.
3. Netmiko 3rd Party Library.
4. Using Python PIP utility to manage Python Libaries.

Lab – Perform configuration change on IOS Routers using Telnetlib.
Lab – Perform IOS Automation using Netmiko Library.

Lab – Configure multiple IOS devices using Python Looping & Conditioning.
Lab – Build python scripts to perform User Authentication with getpass library.

Chapter 7 Application Programming Interfaces (APIs) –

1. Northbound APIs.
2. Southbound APIs.
3. Synchronous Versus Asynchronous APIs.
4. Representational State Transfer (REST) APIs.
5. RESTful API Authentication.
6. Basic Authentication.
7. API Keys.
8. Custom Tokens.
9. Simple Object Access Protocol (SOAP).
10. Remote-Procedure Calls (RPCs).

Chapter 8: RESTful API Requests and Responses – Includes Lab

1. API Types.
2. API Access Types.
3. HTTP Basics.
4. Uniform Resource Locator (URL).
5. Method.
6. REST Methods and CRUD.
7. Deep Dive into GET and POST.
8. HTTP Headers.
9. Request Headers.
10. Response Headers.
11. Response Codes.
12. Tools Used When Developing with Webhooks.
13. Sequence Diagrams.
14. REST Constraints.
15. Client/Server.
16. Stateless.
17. Cache.
18. Uniform Interface.
19. Layered System.
20. Code on Demand.
21. REST API Versioning.
22. Pagination.
23. Rate Limiting and Monetization.
24. Rate Limiting on the Client Side.

25. REST Tools.
26. Postman.
27. Curl.
28. HTTPie.
29. Python Requests.
30. REST API Debugging Tools for Developing.
31. APIs.

Lab - Inspect HTTP Protocol Messages.
Lab - Perform a configuration change on an ASA device with the REST API.
Lab - Use Postman for deploying configuration on ASA-OS.

Chapter 9 Model-Driven Programmability – Includes Lab

1. Foundation Topics
2. NETCONF
3. YANG
4. RESTCONF
5. Model-Driven Telemetry

Lab - Deploy Configurations on IOS devices using Telnetlib, Netmiko, Paramiko.
Lab - Perform a configuration change on an IOS XE device with the RESTCONF.
Lab - Perform a configuration change on an IOS XE device with the NETCONF.

Chapter 10 Deploying Applications

1. Application Deployment Models.
2. NIST Definition.
3. Application Deployment Options.
4. Application Deployment Methods.
5. Bare-Metal Application Deployment.
6. Virtualized Applications.
7. Cloud-Native Applications.
8. Containerized Applications.
9. Serverless.
10. DevOps.
11. What Is DevOps?
12. Putting DevOps into Practice.
13. Docker Architecture.
14. Using Docker.
15. Docker Hub.

Chapter 11 Application Security.

1. Identifying Potential Risks.
1) Common Threats and Mitigations.
2) Open Web Application Security Project.
3) Using Nmap for Vulnerability Scanning.
4) Basic Nmap Scan Against an IP Address or a Host.
2. CVE Detection Using Nmap.
3. Protecting Applications.
1) Tiers of Securing and Protecting.
2) Encryption Fundamentals.
3) Public Key Encryption.
4) Data Integrity (One-Way Hash).
5) Digital Signatures.
6) Data Security.
7) Secure Development Methods.
8) Securing Network Devices.
9) Firewalls.
10) Intrusion Detection Systems (IDSs).
11) Intrusion Prevention Systems (IPSs).
12) Domain Name System (DNS).
13) Load Balancing.

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Chapter 12 Infrastructure Automation – Includes Lab

1. Controller Versus Device-Level Management.
2. Infrastructure as Code.
3. Continuous Integration/Continuous Delivery.
4. Pipelines.
5. Automation Tools 1) Ansible 2) Puppet 3) Chef.

Lab - Creating Ansible .cfg, hosts, and inventory Files.
Lab - Creating and Executing Ansible Playbook for IOS Show commands.
Lab - Backing up IOS Configuration using Ansible.
Lab - Deploying Configuration using Ansible on Network Devices.

Module 2:

Cultural skills:

- Introduction to Soft Skills.
- Effective Communication Skills.
- Interpersonal Skills and Relationship Building.
- Leadership and Management Skills.
- Presentation and Public Speaking Skills.
- Professional Etiquette and Business Ethics.

Next Steps

AimNxt DevNet plays a crucial role in enabling developers to adapt to the changing landscape of networking and infrastructure technologies, making it easier for organizations to adopt automation, programmability, and software-defined networking to meet their business objectives. Whether you're a network engineer, software developer, or IT professional, CCNA DevNet Module provides the tools and resources to enhance your skills and build innovative solutions in the world of networking and IT infrastructure

"Earning a **AimNxt CCNA DevNet Module(ACDM)** is the gateway to a Rewarding and Demanding IT & Software career"

****Course content is dynamic and subject to change without notice****